Teaching a New Product in Arkite (Admin Role)

Phase 1: Pre-requisites & Workstation Verification

Before teaching a new physical product, verify that the workstation hardware and spatial baselines are configured and operational.

1. **Verify Physical Environment**:
    - Ensure the workbench is stable, rigidly positioned, and clear of stray items.
    - Verify lighting conditions are consistent with standard operating production conditions (minimize direct sunbeam exposure or flickering ambient spotlights).
2. **Launch & Login as Administrator**:
    - Launch the Arkite desktop software client.
    - Click the **Menu** icon in the upper-left corner of the window.
    - Select **Log In**.
    - Enter your Administrator credentials (factory default: Arkite 3600 if not updated).
    - Verify that your role indicator in the upper-right title bar displays **Admin**.
3. **Verify Sensor & Projector Calibration**:
    - Navigate to the **Calibration** tab in the main navigation ribbon.
    - Check calibration status for:
        - **3D Sensor (Depth/IR)**: Ensure table plane calibration and ground baseline are valid.
        - **Projector Calibration**: Check that the test grid lines align squarely with the physical table surface and corners.
        - **Arkite Vision / RGB Sensor** (if installed): Ensure extrinsic and intrinsic calibrations match the physical coordinate space of the 3D sensor.
    - If any calibration is misaligned or has been physically nudged, execute recalibration before proceeding.

Phase 2: Project Setup & Draft Mode Access

Arkite requires all structural edits to occur within a modifiable **Draft** version of a project.

1. Click **Menu** →**Projects**.
2. Determine project path:
    - **New Product Project**: Click **New Project**, enter the Project Name, select default language(s), and click **Create**.
    - **Existing Project Line**: Select the existing project from the list.
3. Check the version status in the **Properties** panel on the right:
    - If status is **Production** or **Review**, it is locked for editing.
    - In the ribbon or properties pane, click **Create Draft** / **Add Version**.
    - Set the working status to **Draft**.
4. Open the project in edit mode.

Phase 3: Define the Job and Product Variants

A **Job** represents the complete manufacturing or assembly workflow of the product.

1. Navigate to the **Jobs** tab in the top navigation ribbon.
2. In the Jobs ribbon, click **New Job**.
3. In the right-hand **Properties** panel, configure the following core fields:
    - **Name**: Enter the formal product name (provide translations for all active project languages).
    - **Product Identifier**: Enter the unique Part Number / SKU / Barcode pattern that identifies this product.
    - **Image**: Click the file selector to upload a reference thumbnail image of the finished product.
    - **Expected Cycle Time**: Enter the target takt/cycle time in seconds or minutes.
4. **Configure Job Options** (for products with configurable features):
    - In the Properties panel, locate the **Options** section.
    - Click **Add Option**.
    - Define option name (e.g., Color, DriveType, ModelYear).

- Define permitted values for each option (e.g., Red, Blue, Hybrid, Electric).

5. **Configure Job Variants** (if product has distinct sub-models):
    - In the ribbon, click the **Variants** button.
    - Click **New Variant**.
    - Enter the **Variant Code** (e.g., VAR-001) and **Variant Name**.
    - Assign the specific option value combinations corresponding to this variant.
    - Click **OK**.
6. **Assign Job Groups** (Optional):
    - If organizing a complex assembly line, create a Job Group and drag the new job into it.

Phase 4: Configure the Teaching Canvas & Sensor View

To achieve accurate physical snapshots, adjust the live sensor view before capturing images.

1. Navigate to the **Detections** tab.
2. Select the **Teaching** sub-tab in the central workspace.
3. **Select Viewing Perspective**:
    - **Top-Down (Orthographic)**: Recommended for bin picking, flat components, and tabletop height measurements.
    - **3D Perspective**: Recommended for checking vertical clearance and 3D bounding envelopes.
    - **2D Image**: Standard camera view.
4. **Camera Navigation**:
    - **Scroll Wheel**: Zoom in and out of the workbench area.
    - **Right Mouse Button + Drag**: Pan horizontally and vertically across the field of view.
    - **Left Mouse Button + Drag**: Rotate in 3D perspective mode.
5. **Stream Selection & Mirroring**:
    - Click the **Settings** menu above the stream view.

- Select the optimal sensor stream:
    - **Depth**: Displays 3D spatial height and volumetric distance from the sensor.
    - **IR (Infrared)**: Displays infrared reflectivity (ideal for dark materials, edges, or retro-reflective markers).
- If the operator station is mirrored physically, select **Mirror View** as appropriate.

6. **Sensor Selection**:
    - Open the **Sensor** dropdown menu:
        - Select **3D Sensor** for volumetric objects, bins, and activities.
        - Select **Arkite Vision Sensor** for small parts, high-resolution texture/shape matching, and Quality Checks.
        - Select **RGB Sensor** for color barcoding or standard OCR.
7. **IR Slider Adjustment**:

NOTE

The IR slider adjustment only optimizes the visualization display for the Admin during teaching; it does not alter the underlying algorithmic detection thresholds.

- Adjust the **IR Min/Max Sliders** below the image feed to enhance contrast and edge visibility.

Phase 5: Create Detections

Detections are the virtual sensors monitoring physical actions, tools, parts, and bins on the workstation.

1. **Determine Detection Scope**:
    - **Project-Wide**: Shared across all jobs (e.g., stationary bins, torque screwdrivers, global virtual confirmation buttons).
    - **Job-Specific**: Unique to this product (e.g., specific sub-assembly parts, specialized fixtures).
    - In the left tree, select either the **Project-Wide** folder or select the specific Job in the ribbon, then click the **Job-Specific** folder.
2. Click **New Detection** in the ribbon and select the required detection type:

- Object: Physical components, base assemblies, or workpieces.
- Tool: Manual or powered hand tools (e.g., screwdrivers, calipers).
- Picking Bin: Parts containers and material chutes.
- Activity: Hand movements, insertions, or manual actions within a volumetric space.
- Virtual Button: Projected touch-sensitive zones triggered by hand interruption.
- Quality Check *(Arkite Vision only)*: Inspection for part presence, clip engagement, or screw depth.
- Object with Quality Check *(Arkite Vision only)*: Combined part detection and quality state inspection.
- Text (OCR): Optical character recognition from labels or etched serials.
- Barcode (QR / DataMatrix / 1D): Optical scanning of 1D/2D symbologies.

3. **Configure Core Detection Properties**:
    - In the name dialog, enter a descriptive **Name** for all supported system languages.

IMPORTANT

Use concise, operator-friendly names because they appear directly in the operator instruction display and error banners.

4. **Configure Specialized Properties**:
    - **For Picking Bins**: Click the **Material** property and link to an existing material catalog item, or click **Create New Material**.
    - **For Smart Tools**: If using an integrated DC electric tool, check **Smart Tool** and select the controller protocol (e.g., *Open Protocol / Desoutter / Atlas Copco*).
    - **External Trigger Variable**: If using external sensors (e.g., photoelectric eye or PLC signal), map the boolean variable in the **External Trigger Variable** field.
    - **Pick To Light (DMX)**: If using physical light bars, assign the corresponding DMX channel and LED segment IDs.

Phase 6: Multi-Model Teaching Sequence

Follow the specific physical teaching protocol based on the detection type:

Type A: Teaching Standard Objects, Tools, Picking Bins & Virtual Buttons

1. In the left tree, select the target detection.
2. Switch to the **Teaching** sub-tab.
3. **Define Working Zone**:
    - In the live viewport, drag the corners of the white dashed bounding box to encompass **only** the target area where the object or hand will appear.
    - Keep the zone tight to prevent false positives from adjacent movements.
    - *(Exception)*: For tools with an Arkite Vision license capable of dynamic location tracking, set the bounding rectangle to cover the full physical table surface where the tool could be placed.
4. **Capture "Without" Snapshot**:
    - Completely remove the part, tool, or operator hand from the defined working zone.
    - Ensure the background is clear.
    - Click the **Without...** button.
    - Verify that the snapshot thumbnail displays the clean empty background.
5. **Capture "With" Snapshot**:
    - Place the actual physical part, tool, or bin squarely into its target position.
    - *(For Virtual Button)*: Position an operator's closed fist 50–100 mm above the surface inside the zone.
    - Click the **With...** button.
    - Verify that the snapshot thumbnail clearly captures the placed object.
6. **Set Teaching Parameters**:
    - **Detection Mode**:
        - Automatic (Default, uses intelligent depth/IR combination).
        - Depth (Relies strictly on height above table plane).

        - IR (Relies on infrared reflection).
        - Reflective Tape (IR) (For retro-reflective markers).
        - Infrared Edges (For high-contrast edge boundaries).
        - Shape Based *(Vision)* (Matches contour geometry regardless of color).
        - Texture Based *(Vision)* (Matches detailed surface graphics/textures).
        - Histogram Based *(Vision)* (Matches color distribution).
    - **Position Tolerance**:
        - Wide (Allows placement deviation up to 2.0× component size).
        - Narrow (1.5× component size; recommended default for tools).
        - Accurate (1.2× component size).
        - Exact (1.0× component size; recommended default for fitted parts).
    - **Area of Interest (AOI)**:
        - Set to Automatic by default.
        - If adjacent fixtures cause interference, select Manual →click **Specify...** →use the **Add/Remove Mask Brush** to paint the exact component geometry →click **OK**.
    - **Verify Position** *(Vision Only)*: Check this box to validate (X, Y)coordinates and project directional alignment arrows for the operator.
    - **Verify Orientation** *(Vision Only)*:
        - Check this box if the part must be oriented at a specific angular rotation.
        - For asymmetrical parts, take multiple "With" snapshots at permitted rotational angles. Click the lower-left corner of each snapshot to specify the expected angle.
    - **Adjust Projection**: Ensure this checkbox is checked for initial teaching so Arkite automatically generates projection bounding boxes. Uncheck only if you have already customized projection vectors.
7. Click the **Teach** button.

8. **Verify Result**: Confirm that a bounding box appears around the object with coordinate dimensions matching the physical part.

Type B: Teaching Activities (Hand Actions / Push-Ins / Assembly Moves)

1. Select the **Activity** detection in the tree.
2. In the **Teaching** sub-tab, drag the white dashed bounding rectangle to span the spatial volume where the operator's hands will execute the task.
3. Configure Teaching settings:
    - **Detection Mode**: Select Depth (standard) or Reflective Tape (IR) (if operators wear reflective wristbands/gloves).
    - Check **Adjust Projection**.
4. Click **Teach**.
5. The **Teaching Activity** modal dialog appears with a countdown.
6. **Perform the Physical Action**:
    - Move hands into the zone and execute the exact assembly action at realistic production speed.
    - Retract hands.
7. Click **OK** on the dialog.
8. Verify that the system generates an activity detection box corresponding to the dynamic path.

Type C: Teaching Quality Checks & Objects with Quality Checks (Vision)

1. Select the **Quality Check** or **Object with Quality Check** detection.
2. Position the working zone rectangle tightly over the inspection target (e.g., bolt head, latch, gasket, connector).
3. **Capture Class Snapshots**:
    - **For Quality Check**:
        - **Expected Result (CORRECT)**: Assemble the component properly →Click **Expected Result**.

        - **Bad Result (WRONG)**: Assemble the component improperly (e.g., misaligned, partially inserted) →Click **Bad Result**.
        - **Missing (MISSING)**: Remove the component entirely →Click **Missing**.
    - **For Object with Quality Check**:
        - Capture Without, Expected Result, and Bad Result.
4. **Add Custom Classes** (Optional):
    - In the Properties panel under **Quality Check Classes**, click the **+** icon.
    - Enter Class Name (e.g., StrippedScrew, WrongColorClip).
    - Assign Classification Type: CORRECT, WRONG, or MISSING.
    - Take the corresponding snapshot.
5. **Configure Vision Algorithmic Properties**:
    - **Detection Mode**: Automatic, Shape Based, Texture Based, or Histogram Based.
    - **Position Tolerance**: Set to Exact (standard for quality inspections).
    - **Exposure Mode**: Select Automatic / HDR (standard), or lock to Dark, Normal, or Bright if inspecting reflective metal or deep cavities.
    - **Adjust Occlusion Box**: Keep checked to automatically generate an occluder box from the 3D sensor.
6. Click **Teach**.
7. Confirm that both the detection box and 3D occluder box are generated.

Type D: Teaching Text (OCR) and Barcodes (1D/2D)

1. Select the **Text (OCR)** or **Barcode** detection.
2. Frame the white working zone over the expected label scanning zone.
3. Position the physical sample code or label inside the zone.
4. Click **Add Expected Result**.
5. In the Properties panel, configure:
    - **Variable for Result**: Select or create a string variable (e.g., Var_ScannedSerial) to receive the decoded value.

- **Validate Code/Text**: Check this box if the barcode must match specific rules.
- **Conditions**: Add comparison conditions (e.g., Starts With "ENG-" or Regex match).
- *(For OCR)*: Check **Ignore Whitespace** and **Ignore Casing** to avoid false rejections.

6. Click **Teach**.
7. Verify that the decoded value displays in the preview window.

Phase 7: Individual Detection Testing & Reteaching

Before sequencing jobs, validate each detection independently in live mode:

1. **Test Detection States**:
   - Place the physical component into the station →Watch the detection state toggle to **ON** (Green indicator).
   - Remove the component →Verify state toggles to **OFF** (Grey indicator).
   - Perform a hand pick from the Picking Bin →Verify momentary trigger state.
   - Touch/interrupt the Virtual Button →Verify button trigger registers.
   - Introduce a defective part into the Quality Check zone →Verify classification shifts to **WRONG**.
2. **Execute Reteaching (if validation fails)**:
   - If a detection fails to trigger reliably or flickers:
     - Return to the **Teaching** sub-tab.
     - Clear the snapshots by right-clicking the thumbnail and selecting **Clear**.
     - Re-capture fresh **Without** and **With** snapshots under stable lighting.
     - Adjust **Detection Mode** (e.g., switch from Depth to IR if object is thin or dark).
     - Click **Teach** to overwrite the model.
   - To completely reset a corrupt detection: Right-click detection in tree →**Clear Teaching**.

Phase 8: Box Finetuning & Algorithmic Parameters

When automatic teaching requires physical boundary adjustments, use the **Finetune** workspace.

1. Select the detection in the tree and switch to the **Finetune** sub-tab.
2. In the central viewer, select the generated detection box.
3. Click the **Edit** button above the viewport.
4. **Boundary Geometry Adjustment**:
    - Click and drag the square corner/angular points to re-shape the boundary.
    - **Add Edge Point**: Right-click on a line boundary →Select **Insert Point**.
    - **Delete Edge Point**: Right-click an unwanted point →Select **Remove Point**.
5. **Configure Algorithmic Properties** (in the right-hand panel):

Finetune Property Group	Parameter	Recommended Production Value / Action
Z-Coordinates (Height)	Base Z	Set distance above the table surface where detection starts (e.g., 10 mm to ignore table dust).
	Height	Total vertical height of the detection volume (e.g., 80 mm to encompass the part).
Stream & Image Processing	Use IR	Unchecked = 3D Depth; Checked = Infrared reflectance stream.
	Image Processing	Choose Background Subtraction (standard) or Edge Detection (best for flat stamps).
IR Intensity Thresholds	Min IR Intensity	Minimum pixel brightness threshold ($0 - 255$).
	Max IR Intensity	Maximum pixel brightness threshold ($0 - 255$).

Finetune Property Group	Parameter	Recommended Production Value / Action
Detection Rates	Min Fill Rate %	Minimum percentage of pixels within the box required to toggle detection ON (typically $40\% - 60\%$).
	Similarity %	For Vision: Minimum template matching score (typically $\geq 75\%$).
Debounce & Timing	Time to ON	Milliseconds the condition must persist before triggering ON (e.g., 150 ms prevents false flickers).
	Time to OFF	Milliseconds the condition must persist before toggling OFF (e.g., 200 ms).
Occlusion Handling	Occlusion Check	Checked: Prevents hand or arm shadow from falsely counting as part presence.
	Min Occlusion Rate	Minimum occlusion percentage to trigger an occlusion block (typically 30%).
Vision Tolerances	Position X/Y Tolerance	Allowed physical deviation in millimeters (e.g., ± 5 mm).
	Orientation Tolerance	Allowed angular deviation in degrees (e.g., $\pm 10^\circ$).
	Symmetric Orientations	Check for symmetrical parts (e.g., 90°or 180°symmetry) so reverse placement is valid.

6. Click **Save** / Apply in the ribbon.

Phase 9: Visualization & Projection Mapping

Configure the projected augmented-reality cues displayed on the table surface and operator screens.

1. Select the detection in the left tree.
2. Select the **Projector** sub-tab (for table projection) or **Screen** sub-tab (for operator monitors).
3. Click **New Visualization** in the ribbon:

- Draw Shape: Outlines, circles, rectangles, arrows, or polygons projected directly onto parts/bins.
- Text: On-screen or projected text labels (multi-language enabled).
- Image: PNG/JPG graphic overlays (e.g., wiring diagrams, assembly pictograms).
- Video: Animated instructional loops.

4. **Position & Scale Visualization**:
   - Click and drag the shape handles in the viewport to surround the target detection.
   - Use the magnetic alignment guidelines to snap flush with detection box edges.
5. **Configure Visibility & Timing Trigger States**: In the right-hand **Properties** panel under **Visibility**:
   - **Mode**: Set to On, Blink, or Based on Variable.
   - **Blink Interval**: Set frequency (e.g., 500 ms) for urgent warnings or pick indications.
   - **Trigger States** (Check the specific operating states when this visual must be illuminated):
     - Place: Displayed when operator must put the part down.
     - Take: Displayed when operator must remove the part.
     - Pick: Displayed on picking bins for the active material.
     - Picking Error: Highlighted red if operator reaches into the wrong bin.
     - Press: Highlighted on virtual buttons waiting to be pressed.
     - Perform: Highlighted during activity execution.
     - Shadowboard: Displayed at the tool resting location while the tool is in use.
     - Error (out-of-step): Highlighted if touched out of sequence.
     - Orientation Error: Projected warning arrow if part is placed backwards.
     - Position Error: Projected warning if part is off-center.

6. **Configure Aesthetics**:
    - Set stroke color, fill transparency, line thickness, and font size.
7. **Lock Visualization**:

TIP

Locking visualizations prevents accidental cursor repositioning during future step-sequencing passes.

- Right-click the finished shape →Select **Lock**.

Phase 10: Adding & Structuring Assembly Steps (Core Architecture)

Steps are the fundamental building blocks of the assembly process. In this phase, the Admin establishes the structure, hierarchy, loops, and conditional branches for the product assembly.

10.1 Accessing the Steps Workspace

1. In the top navigation ribbon, select the **Jobs** tab.
2. In the ribbon, select the specific **Job** you are configuring from the dropdown.
3. In the sub-navigation row below the ribbon, click the **Steps** tab.

10.2 Adding Steps

1. Under the **Steps** tree pane, click the **New Step** button.
2. A categorized menu appears. You can browse through tabs or use the **Search** field to type the name of the step type.
3. Select the step type to insert it into the active tree location.

10.3 Setting Up Your Step Structure

Arkite provides four structural mechanisms to organize assembly steps logically for operators:

A. Numbered Steps (Logical Operator Sub-Assemblies)

A **Numbered Step** represents a logical milestone from the operator's perspective (e.g., 1. Fix Cap, 1.1 Place Cap, 1.2 Fasten Screws), containing multiple granular detection steps underneath.

1. Click **New Step** →select **Numbered Step**.
2. In the **Properties** pane, enter the step **Name**.

3. Nest sub-actions inside by dragging detection/material steps under the Numbered Step.
4. *Display Table of Contents*: On the operator's display screen, add a **Numbered Steps List** visualization widget. This acts as an interactive progress table of contents, highlighting the current subassembly.

B. Composite Steps (Execution Order, Parallelism & Loops)

A **Composite Step** bundles two or more child steps, controlling execution order, concurrency, and loop repetitions.

1. Click **New Step** →select **Composite Step** (or select multiple existing steps and click the Group/Composite button).
2. Configure **Properties**:
    - **Name**: Enter a descriptive name.
    - **Child Step Order**: Select the execution logic icon:
        - Sequential Order: Steps execute strictly one after another in order of appearance (Default).
        - Any Order *(Validate & Vision licenses)*: Steps can be executed in any sequence by the operator, but **all** must be completed before advancing.
        - One Arbitrary Step: Completing **any single one** of the child steps satisfies the composite step and advances the job.
        - All Steps Simultaneously *(Validate & Vision licenses)*: All child steps must be executed at the **exact same instant** (e.g., verifying all tools are present at cycle start, or requiring two-hand palm buttons pressed concurrently for safety).
    - **Child Step Instruction Separator**:
        - When using Any Order or All Steps Simultaneously, choose how instructions are merged: Auto (defaults to "and"), And, or Or (for arbitrary steps).
        - Example: If child steps are *Take Screwdriver* and *Pick Screw*, the operator banner automatically displays *"Take Screwdriver and Pick Screw"*. Alternatively, enter a custom instruction on the composite step level.
    - **Repetition (Loops)**:

- None: Runs once and moves to the next step.
- Repeat: Repeats child steps in an endless loop (must be coupled with an exit **Condition**).
- Repeat number of times: Prompts for **From Index** (default 1) and **To Index** (or assign integer variables like {{LoopCount}}). Saves the index into a **Variable for Index**.
- Repeat for each element: Loops through a collection:
    - **List Type**: Choose Json Array or List with Separator (e.g., , or ;).
    - **List of Elements to Loop Over**: Point to string variable containing the data.
    - **Variable for Current Element**: Stores current iteration value.
    - **Variable for Count**: Stores total element count.
    - **Variable for Index**: Stores active iteration index.

C. The Switch Job Option Step (Variant Branching)

Used when different product options (e.g., Fuel Type = Petrol, Diesel, or Electric) require different physical assembly branches.

1. Click **New Step** →select **Switch Job Option**.
2. In Properties, select the target **Job Option**.
3. Arkite automatically creates a distinct sub-branch in the tree for every possible option value.
4. **Merge Cases**: If Petrol and Diesel share the exact same steps, drag the Diesel option into the Petrol case to merge their step branches.
5. Under each branch, add the specific assembly steps required for that option.

D. The Switch Step (Variable-Driven Branching)

Directs process execution down different sub-paths based on the real-time value of an Arkite variable (e.g., Torque_Result == "Within_Tolerance" vs. "Too_High" vs. "Too_Low").

1. Click **New Step** →select **Switch Step**.
2. Configure Properties:
    - **Variable for Switch**: Select the monitoring variable.

- **Switch Cases**: Click + to add target value cases (e.g., Pass, Fail, Rework).
- **Add Default Case**: Check this box to handle any unassigned values.

3. Under each generated subpath, insert the corrective or downstream steps.

E. The Check Step with Rework Flows (Vision License Only)

Used with Quality Check, Object with Quality Check, Barcode, and Text (OCR) detections.

1. Click **New Step** →select **Check Quality** (or link detection).
2. Arkite automatically generates **Rework Flow** substeps for every detection class:
   - For Quality Checks: Substeps for Expected Result (CORRECT), Bad Result (WRONG), and Missing (MISSING).
   - For Barcode/Text: Substeps for Expected Result, Bad Result, and Missing.
3. In each Rework Flow substep:
   - Add corrective steps (e.g., unfasten, replace, or rescanning instructions).
   - In Properties for WRONG or MISSING, enter the **Error Message** to project in bright red on table and screen.
4. **Looping Behavior**:
   - If evaluation is CORRECT, the flow executes and immediately advances to the next assembly step.
   - If evaluation is WRONG or MISSING, the system executes the rework steps and **automatically re-evaluates the Quality Detection** in a continuous loop until the Expected Result is achieved.

10.4 Organizing and Managing Steps in the Tree

- **Reorder Steps**: Click and drag any step up or down in the tree.
- **Duplicate Steps**: Right-click a step →select **Duplicate**.
- **Delete Steps**: Select the step →press Delete on keyboard, or right-click →**Delete**.
- **Move to Job**: Right-click step →**Move** →**To Job** →select destination job.
- **Move to Process**: Right-click step →**Move** →**To Process** →select target background process.
- **Tree View Control**: Use the **Expand All** and **Collapse All** icons at the top of the tree to inspect nested child structures.